4. VACUUM CLEANER AGENT

```
def vacuum_agent(location, status_A, status_B):
    if location == "A" and status_A == "Dirty":
        return "SUCK"
    elif location == "B" and status_B == "Dirty":
        return "SUCK"
    elif status_A == "Dirty":
        return "LEFT"
    elif status_B == "Dirty":
        return "RIGHT"
    else:
        return "NO-OP"



def vacuum_simulation(location, status_A, status_B):
    print("\nInitial State:")
    print(f"Location: {location}, Room A: {status_A}, Room B: {status_B}")

    while True:
        action = vacuum_agent(location, status_A, status_B)

        if action == "NO-OP":
            break

        if action == "SUCK":
            if location == "A":
                status_A = "Clean"
            else:
                status_B = "Clean"
```

```
    elif action == "LEFT":
        location = "A"

    elif action == "RIGHT":
        location = "B"

    print(f"Action: {action}")
    print(f"Location: {location}, Room A: {status_A}, Room B: {status_B}")

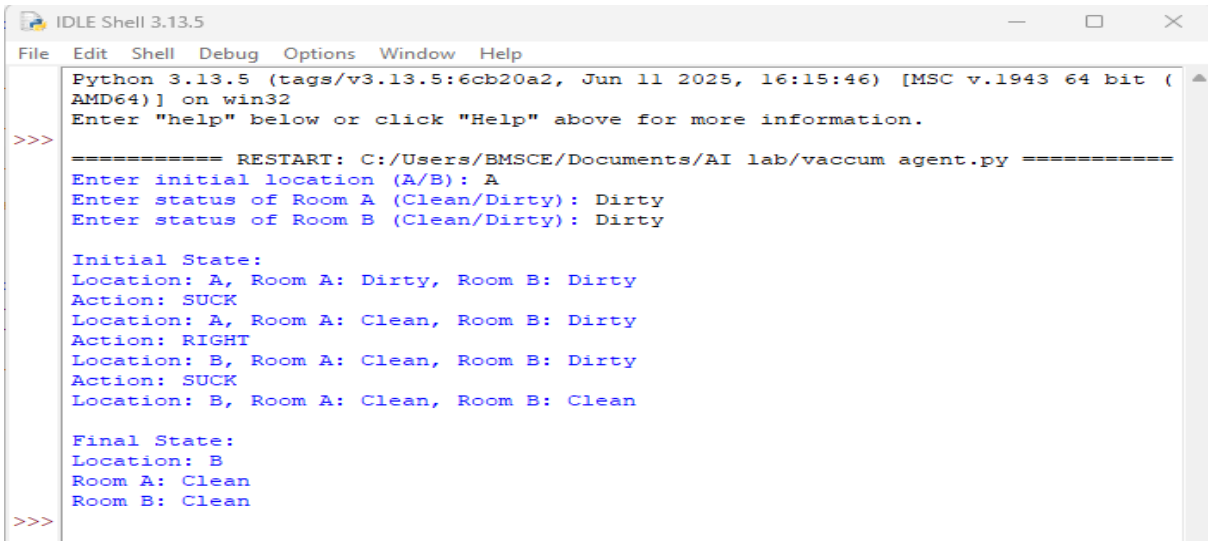
print("\nFinal State:")
print(f"Location: {location}")
print(f"Room A: {status_A}")
print(f"Room B: {status_B}")

location = input("Enter initial location (A/B): ").strip().upper()
status_A = input("Enter status of Room A (Clean/Dirty): ").strip().capitalize()
status_B = input("Enter status of Room B (Clean/Dirty): ").strip().capitalize()

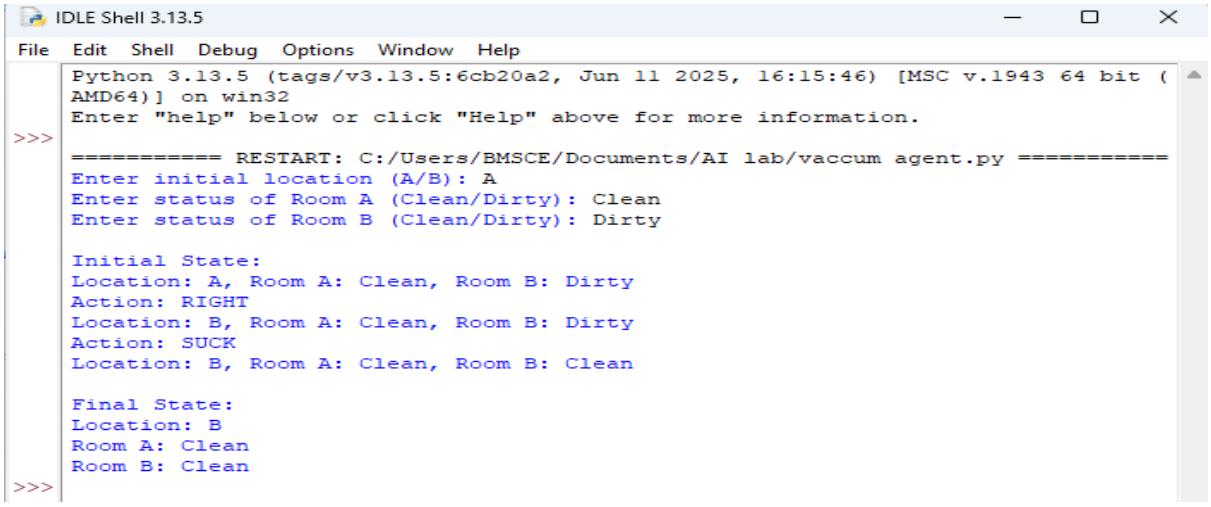
vacuum_simulation(location, status_A, status_B)
```

Output:

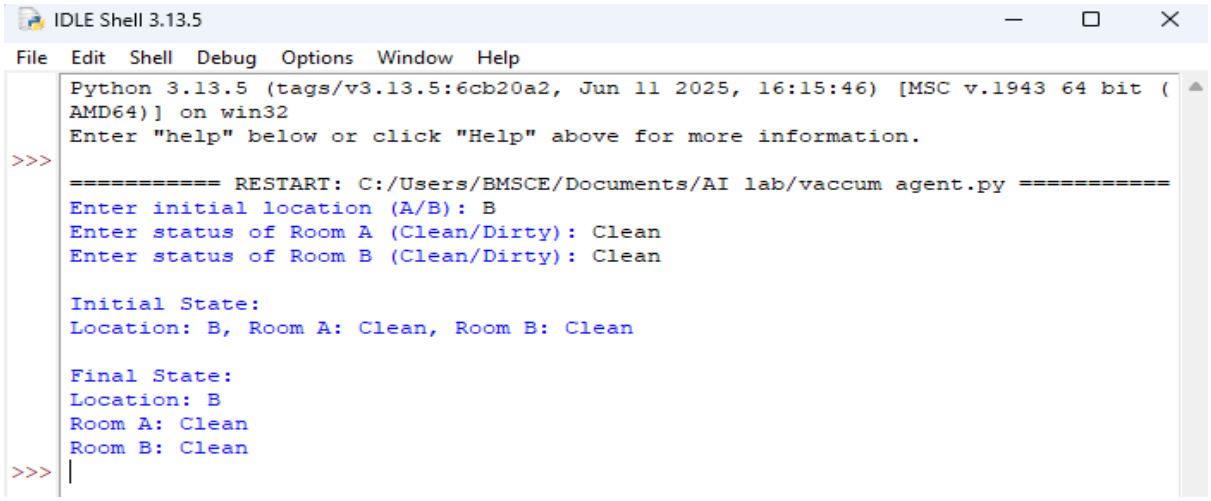
Test case 1 : Both rooms dirty, starting in A



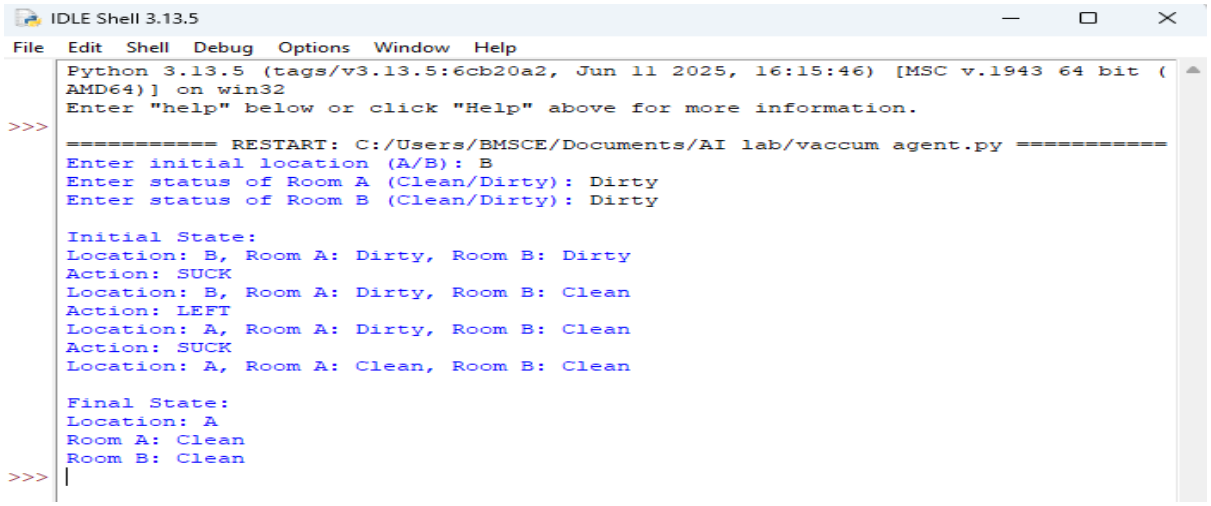
Test case 2 : Room A clean, Room B Dirty, starting in A



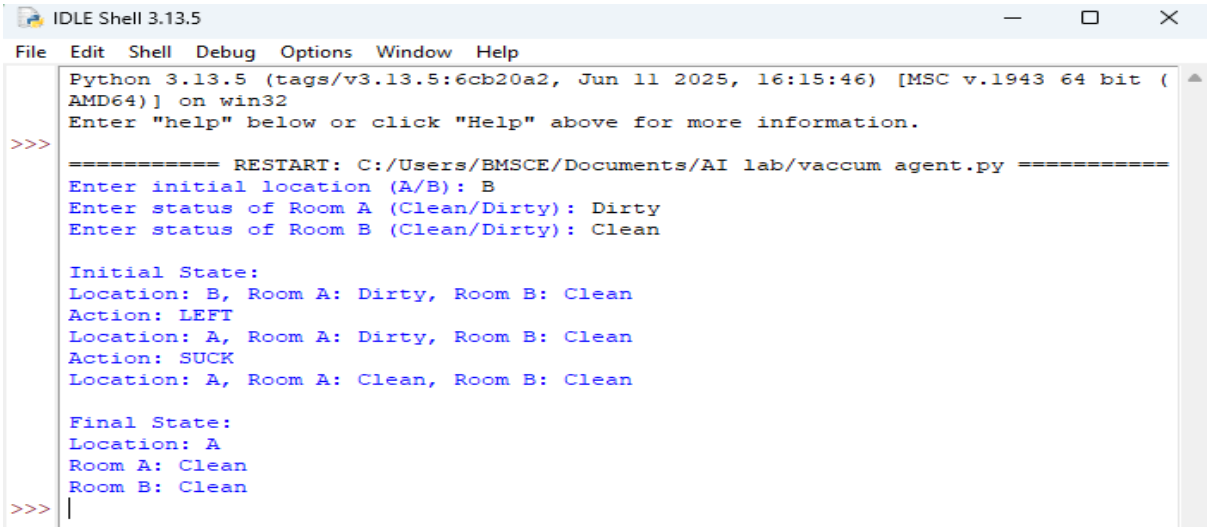
Test case 3: Both Rooms Clean, Starting in B



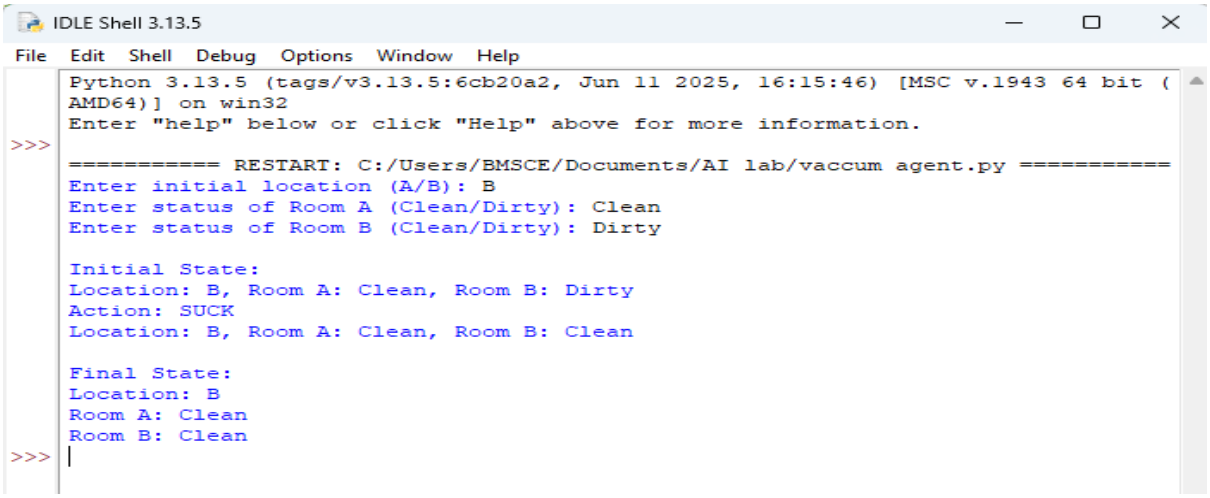
Test case 4: Both Rooms Dirty, starting in B



Test case 5 : Room A Dirty, Room B clean, starting in B



Test case 6 : Room A Clean, Room B Dirty, starting in B



Test case 7 : Both Rooms Clean, starting in A

